

Nishita Matlani

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in LinkedIn

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🌐 Portfolio

EDUCATION

Northeastern University, Boston, MA – *Master of Science, Information Systems*

July 2025

Relevant Courses: Business Intelligence, Intelligent Systems (GenAI), Machine Learning, Data Science

University of Mumbai, Mumbai, India – *Bachelor of Engineering, Computer Engineering*

June 2023

Relevant Courses: Data Structures, Algorithms, Object Oriented Programming, Financial Analysis

TECHNICAL SKILLS

Languages:	Python, SQL, R, Java, Shell scripting
Databases:	Snowflake, PostgreSQL, MS SQL Server, MySQL, MongoDB, Oracle DB, Neo4j
Frameworks:	NumPy, pandas, Seaborn, Matplotlib, scikit-learn, PyTorch, Spark, Airflow, Kafka
Platforms & Tools:	Docker, Grafana, Git/GitHub, Databricks, DBT, GCP, BigQuery, AWS (Glue, Athena, Redshift)
Machine Learning:	Regression, Classification, Clustering, PCA, Grid Search, Bayesian Optimization

EXPERIENCE

AI Engineering Intern

Jan 2025 – April 2025

Humanitarians AI – Boston, MA

- Developed secure **RESTful APIs** to cut data sync delays by **25%**, enabling real-time clinical analytics by integrating EHR, OpenSearch and lab endpoints with an **AWS** platform, supporting customer-focused solutions via Swagger and Postman
- Designed a HIPAA-compliant generation pipeline to cut query latency by 25%, improving trial feasibility for researchers, by modeling biomedical agentic relationships in **Neo4j** for AI-driven retrieval via **ChromaDB**
- Created an intuitive natural language-to-**SQL** interface to reduce analytics backlog by 20%, empowering clinical staff and non-technical stakeholders to self-serve cohort queries, by leveraging **NLP** for precise **SQL** generation
- Fine-tuned a BERT-based domain embedding model using **contrastive** and **supervised** learning to better capture healthcare-specific jargon, **improving semantic similarity** performance by 28% on clinical benchmarks

Software Engineering Intern

Dec 2022 – June 2023

EPM Consultancy – Mumbai, India

- Aggregated **5M+** e-commerce financial transactions into a data warehouse, cutting manual reconciliation by 60%, by building **ETL pipelines** with AWS Lambda, Amazon S3, and PySpark from 5 sources such as Paytm APIs, ERP
- Boosted financial reporting reliability by 95%, developing **Python validation scripts** to ensure information capture and enforce schema across budget and forecast data, improving month-end close accuracy across 8+ business units
- Built **time series forecasts** using Prophet in Python and **SQL** on **AWS Redshift** for 6 months of sales and revenue data across 20+ SKUs, applying first-principles reasoning to achieve 8% MAPE; insights supported GTM and **supply chain forecasting**
- Reduced reporting latency by 25% for **SEO** and financial insights, delivering **KPI dashboards** using data storytelling via **AWS QuickSight**, managing GitHub code, and tracking JIRA tasks in Agile sprints

Data Analyst Intern

Feb 2022 – Sept 2022

EPM Consultancy – Mumbai, India

- Automated demand and product data workflows in **MS SQL** using stored procedures, triggers, and functions, cutting manual effort by 40% and improving **operational insights into inventory turnover** and customer ordering for supply chain planning
- Supported KPI definition with Revenue Operations; built **Power BI** dashboards (DAX, Power Query, Custom JavaScript) to track fuel, demand shifts, and vendor performance, providing **GTM insights** and reducing reporting cycles by 12%.
- Used **logistic** and **linear regression** in Python (pandas, scikit-learn) and **SQL** to analyze e-commerce and B2B SaaS customer behavior, identifying retention drivers and pricing sensitivity, boosting conversion rates by 12% and repeat purchase frequency
- Supported **A/B testing** and funnel analysis for e-commerce and B2B platforms using **SQL**, **Power BI**, and Tableau, uncovering drop-off points and pricing sensitivity that increased conversion rates by 12% and boosted reorder frequency

PROJECTS

DocQuest – (Airflow, Docling, OpenAI, Pinecone, LangGraph, Snowflake, FastAPI, Docker, HuggingFace API)

- Built a scalable, GenAI-powered pipeline to automate the ingestion and semantic parsing of **10,000+ academic documents**, leveraging **Docling**, **Pinecone**, and **FastAPI** for real-time vector retrieval and intelligent document structuring
- Deployed a modular, LLM-integrated orchestration framework with **LangGraph** and **RAG pipelines**, containerized with **Docker**, deployed on **EC2**, and fortified with automated **CI/CD** workflows for seamless, production-grade pipeline operations

Disaster Response Analyzer – (Python, scikit-learn, XGBoost, NLTK, RandomForest, Flask, S3, RDS, EC2, MLOps)

- Extracted data from **S3/RDS**, implemented advanced NLP cleaning with **MLOps** integration and ensemble models (**Random Forest**, **XGBoost**), achieving an **F1-score of 0.85** and **30% reduction** in inference latency
- Orchestrated **Flask & JavaScript** deployment on **EC2** with **CI/CD** pipeline, implementing **monitoring logging**, and **integration testing** systems for real-time text classification